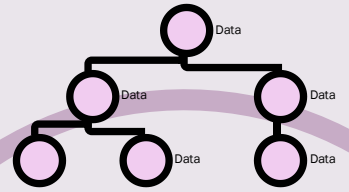
SQL SEARCH ENGINE

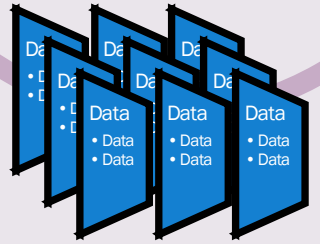
What is it and how does it work?

Record- or contact-based templates



1. Retrieve a set of records or contacts from Clarity or Caboodle.

SQL Query



Ad Hoc Chronicles search

2. Use the results of the SQL query as the base set of records or contacts for a Chronicles search.

3. Display Chronicles data about the resulting records or contacts using PAF columns.

Results

1. Retrieve a set of records or contacts from Clarity or Caboodle.

SQL Query

- Executed on demand from Hyperspace
- Returns a set of Chronicles identifiers
  - *Record or Contacts*
- Does not control the display

Ad Hoc Chronicles search

2. Use the results of the SQL query as the base set of records for a Chronicles search.

- Happens immediately after the SQL query during one report run
- Searches only the Chronicles subset
- Follows all the rules of an Ad Hoc only Workbench template
  - *Base search master file*
  - *Networked parameters*
  - *PAF columns for display*
  - *Real time data*

3. Display Chronicles data about the resulting records or contacts using PAF columns.

Results

- Results are HRN records
- SQL results and Ad Hoc results are not stored separately
- Technically a combination of historical and real time data

3 types of PAF column

Data Item	Extension	SQL Column
• Retrieves display value from INI+Item # • Real time	• Runs extension(s) to display data • Real time	• Retrieves display value from SELECT clause of SQL query • Uses alias to find data (case sensitive) • Not real time

SPECIAL SYNTAX

Passing information between SQL and Chronicles

Write it Down

- Medication dispenses are contacts in the ORD master file. These contacts can only be identified by CDRs in Chronicles. To write a Cogito SQL template that returns a list of medication dispenses, you must use **Clarity**.
- You could not complete this report using **Caboodle**.

Unique identifiers

Records

- In Chronicles
  - *.1 (the record ID)*
- In Clarity
  - **_ID column*
  - *Often primary key*
- In Caboodle
  - **EpicId*
  - *Never primary key*

Contacts

- In Chronicles
  - *Contact Serial Number (item number varies)*
  - *Contact Date Real (CDR, not an item)*
- In Clarity
  - **_CSN*
  - **_DATE_REAL + *_ID*
- In Caboodle
  - **EpicCsn*
  - *Caboodle doesn't store CDRs*

Aliases

- ID
- CSN
- DATE
  - *Not an actual date*
- LINE

Column name	Alias	Column name	Alias
VisitEpicCSN	CSN	ALT_DATE_REAL	DATE
TX_ID	ID	CONT_DAT_REAL	DATE
ORDER_DATE_REAL	DATE	PAT_ENC_CSN_ID	CSN
NDC_CSN_ID	CSN	CNCT_SERIAL_NUM	CSN
MedicationEpicId	ID	ACCOUNT_ID	ID
GuarantorEpicId	ID	LINE	LINE

COGITO SQL PARAMETERS

Rules for Cogito SQL Parameters

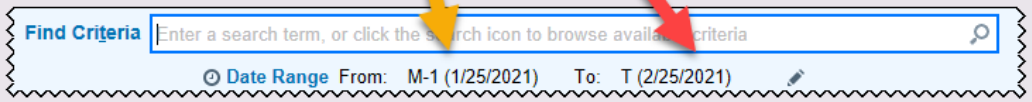
- Searching on multiple values?
  - *Use IN ({{parameter}}) instead of equals (=)*
- Optional parameters must allow for a NULL parameter value
  - *OR NULLIF(COALESCE({{parameter}},N''),N'') is NULL*
- Want to allow report writers to pick their own operators?
  - *Build in Query Template > Parameters only. Do not handle in SQL Query.*
- Want to hard-code a parameter?
  - *Build as normal, but add default value in Query Template > Parameters*
  - *Make required or default/hidden in User Interface > Parameters*

DATE RANGES

```
SELECT
PAT_ENC_CSN_ID      "CSN"

FROM
V_SCHED_APPT appt

WHERE
appt.APPT_STATUS_NAME = 'cancelled'
and
appt.APPT_MADE_DATE < ({{REPORT_END_DT}} + 1)
and
appt.APPT_MADE_DATE >= {{REPORT_START_DATE}}
```

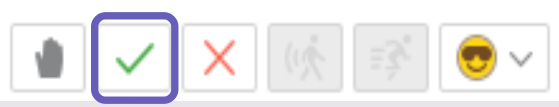


Rules for Date Ranges

- Always use >= {{REPORT_START_DT}}
- Always use < ({{REPORT_END_DT}} + 1) to handle datetimes
- In Caboodle, join to DateDim to turn a DateKey into a DateValue

Exercise 1: Writing a Query for a Cogito SQL Template

- Open SSMS through access.epic.com
- You can find the exercise file under This PC > Exercise Answer Keys > Epic 2018 and later > COG2030v
- When you are finished, click the green check mark in the participants window



NON-CHRONICLES SQL QUERIES

Where we're going, we don't need roads

Search On: “Defined by Search”

- Requirements
  - *Parameters use “Cogito SQL Parameter” property*
  - *Manually assign data type, name, category list or master file*
  - *Manually set display, help, tooltips in User Interface*
- Restrictions
  - *All parameters must be handled by SQL query*
  - *No real time analysis or display*
  - *All columns must be SQL PAFs*
  - *No detail views in results*
  - *No toolbar actions*

Choose a scenario to work on

- Scenario 1: Clarity
- Report on implants used in surgical procedures

- Scenario 2: Caboodle
- Report on hospital accounts used in billing

Ask questions in the Q&A window for trainer assistance

Evaluations at Training.epic.com/evaluations

• © 2021 Epic Systems Corporation. All rights reserved. PROPRIETARY INFORMATION - This item and its contents may not be accessed, used, modified, reproduced, performed, displayed, distributed or disclosed unless and only to the extent expressly authorized by an agreement with Epic. This item is a Commercial Item, as that term is defined at 48 C.F.R. Sec. 2.101. It contains trade secrets and commercial information that are confidential, privileged and exempt from disclosure under the Freedom of Information Act and prohibited from disclosure under the Trade Secrets Act. After Visit Summary, Analyst, App Orchard, ASAP, Beacon, Beaker, BedTime, Bones, Break-the-Glass, Bugsy, Caboodle, Cadence, Canto, Care Everywhere, Charge Router, Chronicles, Clarity, Cogito ergo sum, Cohort, Colleague, Comfort, Community Connect, Cosmos, Cupid, Epic, EpicCare, EpicCare Link, Epicenter, Epic Earth, EpicLink, EpicWeb, Garden Plot, Good Better Best, Grand Central, Haiku, Happy Together, Healthy Planet, Hyperspace, Kaleidoscope, Kit, Limerick, Lucy, Lumens, MyChart, OpTime, OutReach, Patients Like Mine, Phoenix, Powered by Epic, Prelude, Radar, Radiant, Resolute, Revenue Guardian, Rover, Share Everywhere, SmartForms, Sonnet, Stork, System Pulse, Tapestry, Trove, Welcome, Willow, Wisdom, With the Patient at Heart, and WorldWise are registered trademarks, trademarks, or service marks of Epic Systems Corporation in the United States of America and/or other countries. Other company, product, and service names referenced herein may be trademarks or service marks of their respective owners. Patents Notice: www.epic.com/patents.